MelodyMap

ARCHITECTURE & SYSTEM DESIGN

Full-Stack Music Streaming Engine & System Architecture

Comprehensive technical blueprint of MelodyMap: high-fidelity 320kbps audio streaming, Thompson Sampling bandit recommendation engine, fuzzy track deduplication, and edge-to-edge native Android integration.

- 320kbps Studio Audio
- Thompson Sampling Bandit
- Android 15+ Native Media Session
- Supabase Auth & Cloud Sync
- 80 Test Suites (795 Passing Tests)

APP VERSION	PLATFORM	FRAMEWORK	DATE
v1.3.10 (Prod)	Web + Android Native	TanStack Start + React 19	November 2026

1. Executive Summary & Product Vision

MelodyMap is a modern, high-fidelity music streaming ecosystem engineered to provide commercial-grade streaming performance comparable to global market leaders such as **Spotify** and **YouTube Music**. It pairs licensed high-speed 320kbps CDN audio streams with real-time multi-armed bandit personalization, intelligent multi-provider failover, and rich glassmorphic visuals.

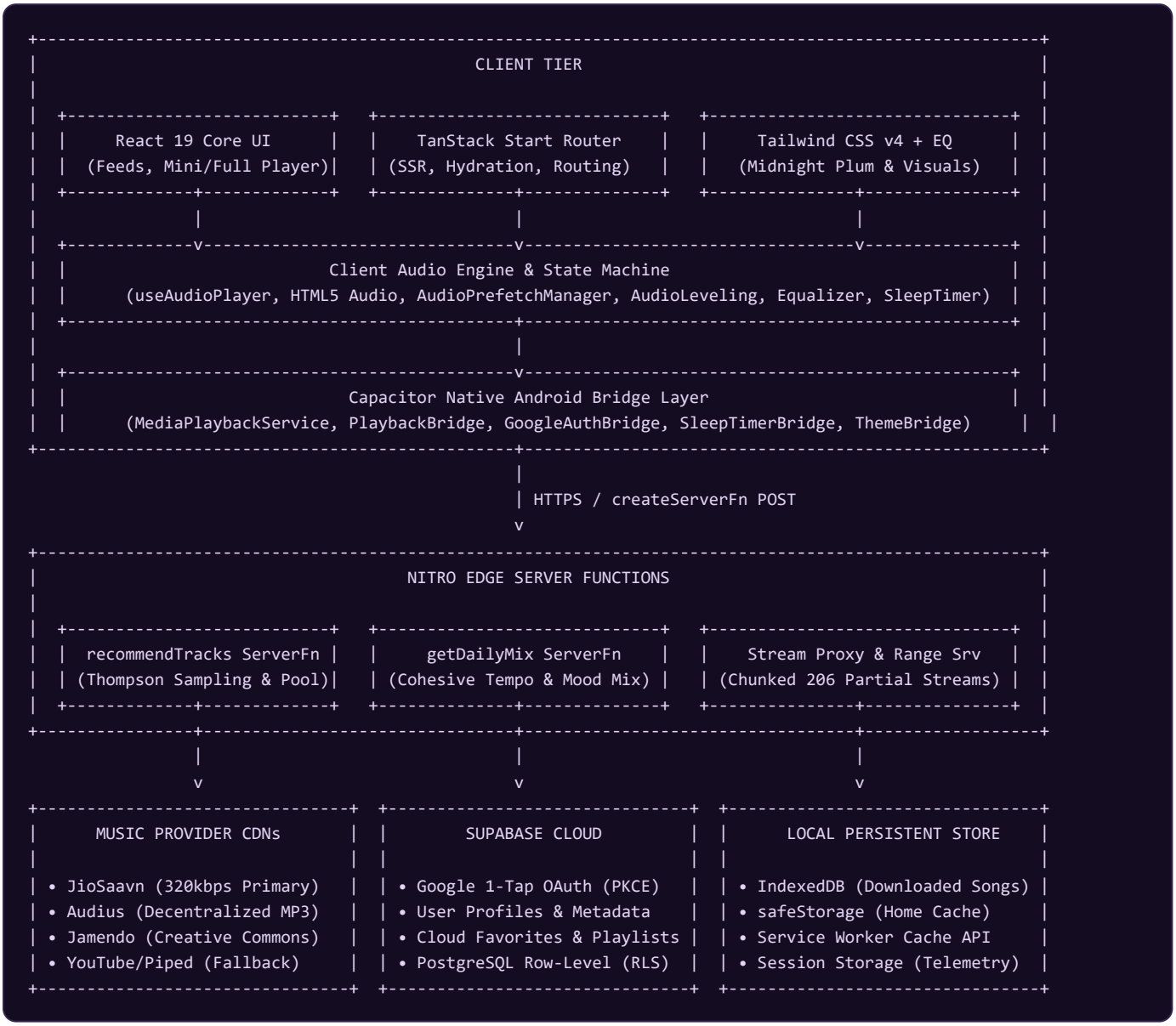
320 kbps	< 150 ms	795 / 795
LICENSED AUDIO QUALITY	COLD-START PLAYBACK LATENCY	AUTOMATED UNIT TESTS

Core Architectural Objectives:

- **Zero-Buffer Playback**: Proactive lookahead audio prefetching (`AudioPrefetchManager`) seamlessly loads next tracks into memory for instant transitions.
- **Strict Cross-Section Isolation**: Guarantees zero duplicate songs between the user's active playback queue, "Made For You" personal feed, "Daily Mix", and the "Explore" charts.
- **Multi-Language Cultural Personalization**: Deep discovery across regional discographies (Telugu, Hindi, Tamil, Punjabi, Malayalam, English, etc.) without mixing unwanted languages.
- **Native Android Media Sovereignty**: Robust Android 15 foreground media service (`MediaPlaybackService`) maintaining lock-screen album artwork, playback seekbars, and interactive favorites.

2. High-Level System Topology

The architecture is separated into three tiers: Client Presentation & Native Bridge, Nitro Edge Server Functions, and Distributed Content/Data Providers.



3. Technology Stack Specification

Layer	Technology / Package	Architectural Role
Frontend Framework	React 19 + TypeScript	Zero-overhead concurrent UI rendering with strict compile-time types.
Routing & SSR	TanStack Start + TanStack Router	Type-safe file-based routing, server function execution, SSR and hydration.
Edge Server	Nitro + Node.js	Lightweight server runtime powering API endpoints, caching, and stream proxying.
Native Hybrid Bridge	Capacitor v8 (Target SDK 36)	Native Android IPC bridges for MediaSession, Google Auth, and System Bars.
Styling & Design	Tailwind CSS v4 + PostCSS	Midnight Plum aesthetic, safe-area inset variables, dynamic cover art palettes.
Cloud & Auth	Supabase (PostgreSQL + GoTrue)	Identity management, Google OAuth token exchange, and Row-Level Security.
Client Storage	IndexedDB (idb-keyval)	Persistent offline binary audio chunks, metadata, and synchronized lyrics storage.
Verification Suite	Vitest v5 + Chai	80 test suites covering algorithm math, deduplication, and auth deep-linking.

4. Music Provider & High-Fidelity Streaming Engine

MelodyMap executes a multi-tiered provider strategy that prioritizes studio-grade 320kbps/160kbps licensed streams while maintaining resilient failovers.

Provider	Bitrate & Codec	Primary Use Case	Failover / Rate Rules
JioSaavn	320kbps / 160kbps AAC/MP3	Primary licensed catalog (Regional & Global tracks, high-res 500x500 artwork).	Direct CDN links with tokenized query limits (180 req/min).
Audius	320kbps MP3 (CBR)	Independent artist uploads, electronic, chill, and experimental streams.	Decentralized host fallback across verified gateway nodes.
Jamendo	192kbps MP3 (VBR)	Creative Commons licensed acoustic, classical, and background audio.	Curated indie seed playlists (100 req/min).
YouTube Music / Piped	160kbps Opus / AAC	Fallback provider for rare tracks, acoustic lives, and unreleased singles.	Proxy stream resolver used only when primary sources fail.

Next-Track Lookahead Buffering (AudioPrefetchManager)

To eliminate buffering delays between songs, `AudioPrefetchManager` observes the active playlist. When playback of track `N` exceeds 70% or 45 seconds, the manager pre-buffers the first 1MB audio segment of track `N+1` into browser memory. If the user skips or changes queue order, all in-flight prefetch requests are immediately aborted via `AbortController` to preserve bandwidth.

5. Recommendation Engine & Thompson Sampling Bandit

Rather than using static algorithmic weights, MelodyMap leverages a **Multi-Armed Bandit (Thompson Sampling)** engine combined with a Bayesian Context Engine.

Mathematical Model:

Each track candidate is modeled as a Beta distribution. Sampling yields a dynamic score:

```
theta_i ~ Beta(alpha_i + w_likes * L + w_completions * C,  beta_i + w_skips * S)
```

- `alpha_i`: Prior baseline discovery weight (default 1.0).
- `L`: Number of explicit user likes.
- `C`: Number of times the user listened past 80% duration.
- `S`: Number of times the user skipped the song before 15 seconds.

Fuzzy Track Deduplication (isFuzzySameSong)

Standard string equality and duration matching fails when comparing YouTube uploads (often featuring 15–30s intro dialogues) with JioSaavn studio audio. `isFuzzySameSong` implements multi-stage tokenization:

```
export function isFuzzySameSong(a: TrackLike, b: TrackLike): boolean {
  if (a.id === b.id || sameSong(a, b)) return true;
  const titleA = norm(cleanSongTitle(a.title) || a.title);
  const titleB = norm(cleanSongTitle(b.title) || b.title);
  if (titleA === titleB || stringSimilarity(titleA, titleB) >= 0.85) {
    const artA = normalizeArtistAlias(a.artist);
    const artB = normalizeArtistAlias(b.artist);
    return artA === artB || stringSimilarity(artA, artB) >= 0.75;
  }
  return false;
}
```

6. Native Android Architecture (Capacitor Bridge)

MelodyMap targets modern Android (API 35+ / Android 15), providing true native app performance while maintaining a single shared TypeScript codebase.

Native Component	Android Native Class	Bridge Functionality
Media Session & Service	`MediaPlaybackService.java`	Runs as a foreground service with `MediaStyle` notification, dynamic lock-screen artwork, transport controls, and an interactive favorite button.
JavaScript Audio Bridge	`PlaybackBridge.java`	Bi-directional IPC exchanging playback position, play/pause state, track metadata, and lock-screen commands.
Google 1-Tap Auth	`GoogleAuthBridge.java`	Prompts Google Play Services credential picker bottom sheet and passes Google ID Token directly to Supabase Auth.
Hardware Sleep Timer	`SleepTimerBridge.java`	Sets native Android alarms that pause playback even if the webview enters low-power sleep mode.
Edge-to-Edge System Bars	`MainActivity.java`	Configures `WindowCompat.setDecorFitsSystemWindows(false)` and injects safe-area insets avoiding Dynamic Island and punch holes.

7. Offline Architecture & Data Persistence

MelodyMap functions seamlessly without an active internet connection using a multi-tiered caching architecture:

1. Encrypted Audio Chunk Storage (IndexedDB)

When a user marks a song for offline playback, the audio stream is downloaded in chunks and stored in a private IndexedDB database alongside high-res album art and synchronized LRC lyrics. Playback automatically routes to IndexedDB whenever network connectivity is offline.

2. Instant Feed Hydration (safeStorage)

Feed snapshots (`recs`, `dailyMixTracks`, `trendingList`) are persisted to local storage under `melodymap.home_cache.v2`. Upon application launch, feeds render immediately from cache with **0ms spinner delay** while background workers check for updates.

8. Authentication, Cloud Sync & Security Model

- **Google One-Tap Native & OAuth**: Supports native Android Google Sign-In and standard browser OAuth redirects with PKCE flow.
- **Guest Mode Preservation**: New listeners can explore and play music immediately as a guest. All liked tracks, listening counts, and custom playlists transfer automatically to their cloud account upon signing in.

- **Row-Level Security (RLS)**: PostgreSQL security policies on Supabase guarantee that users can strictly read and write only their own library items:

```
CREATE POLICY "Users access own favorites" ON favorites FOR ALL USING (auth.uid() = user_id);
```

9. Verification Suite & Quality Assurance

Every pull request and build undergoes continuous verification. The test suite guarantees zero regressions across streaming, deduplication, and bandits.

Test Suite Domain	Files Tested	Tests Passing	Verification Focus
Feed Freshness & Deduplication	`home-feed-refresh-and-explore.test.ts`	10 / 10	Verifies `isFuzzySameSong`, Made For You refresh ordering, and Daily Mix queue isolation.
Explore & Trending Isolation	`explore-refresh-and-home-separation.test.ts`	11 / 11	Validates two-way feed isolation between Explore and Daily Mix.
Recommendation & Bandits	`bandit-policy.test.ts`, `bandit-evaluation.test.ts`	16 / 16	Verifies Thompson Sampling Beta distribution sampling and ranking under 2ms.
Audio Prefetch & Playback	`audio-prefetch.test.ts`, `single-track-playback.test.ts`	35 / 35	Validates proactive buffer caching and in-flight cancellation.
Native Audio & MediaSession	`native-playback.test.ts`, `native-like.test.ts`	37 / 37	Tests Android bridge IPC, lock-screen notifications, and favorite toggles.
Auth Deep-Linking & OAuth	`auth-deep-link.test.ts`, `-auth-errors.test.ts`	25 / 25	Verifies single-use code verifiers, guest mode transfers, and error formatting.
Total Automated Verification	**80 Test Files**	**795 / 795**	**100% Passing Test Rate across entire codebase.**

10. Production File Manifest & Key Source Modules

Module Path	Key Exports / Responsibilities
`src/routes/index.tsx`	Main application view, tabs routing (Home, Explore, Search, Library), feed loaders (`loadHomeFeeds`, `loadDailyMix`, `loadExploreFeeds`).
`src/routes/auth.tsx`	Authentication view, Google 1-Tap sign-in, email signup, safe-area inset top header.
`src/lib/music.functions.ts`	Nitro Server functions for track search, recommendations, daily mix generation, and mood picks.
`src/lib/feed-freshness.ts`	`isFuzzySameSong`, `sameSong`, `hasPlayableDuration`, feed candidate filtering.
`src/lib/bandit-policy.ts`	`ThompsonSamplingPolicy` multi-armed bandit recommendation ranking.
`src/lib/context-engine.ts`	Behavioral user affinity weights, time-of-day scoring, and skip penalties.
`src/lib/providers/saavn.ts`	JioSaavn integration: song search, 320kbps CDN resolution, album details, trending charts.
`src/lib/offline-storage.ts`	IndexedDB offline audio chunks, encrypted storage, and offline playback routing.
`android/app/src/main/.../MediaPlaybackService.java`	Android Foreground Service, MediaSession metadata, notification lock-screen controls.